

3D PRINTED SILICONE

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MATERIAL BENEFITS OF SILICONE

Silicone offers a unique combination of properties not found together in other substances. Its molecular structure provides great flexibility and elastic recovery, allowing it to bend, stretch, compress, and rebound without cracking or losing form, which is ideal for applications requiring dynamic movement. It also demonstrates reliable thermal stability across a wide temperature range, remaining flexible at low temperatures without becoming overly soft at high temperatures. Furthermore, silicone can be formulated for biocompatibility, meeting standards such as cytotoxicity, which makes it a trusted material for skin-contact applications in consumer wearables and medical devices.

Applications:

- Wearables & Consumer Products – Ergonomic watch bands, custom grips, and soft-touch enclosures.
- Medical Devices & Healthcare – Patient-specific prosthetic liners, surgical training models, and biocompatible seals.
- Automotive & Transportation – Custom gaskets and vibration dampeners.

WHY IS 3D PRINTING SILICONE GAINING RELEVANCE?

Growing demand for personalized medical devices (prosthetics, implants, hearing aids) has driven the need for manufacturing methods capable of producing patient-matched parts. Advances in printing processes and material formulations now allow silicone to be additively manufactured.



Swiss company MotionTech creates the customized Your Liner, which are 3D printed silicone prosthetic liners, from individual patient scan data (Source: MotionTech)

PROCESS

Silicone is different from standard 3D printing materials. It can't be melted down like plastic filament, and pure silicone doesn't harden when exposed to UV light. Because of this, it cannot be printed on a normal 3D printer. Instead, companies have created special machines and methods to work around these challenges.

These new processes allow liquid or paste-like silicone to be laid down layer by layer and then solidified using heat, specialized UV formulas, or chemical reactions. This makes it possible to build flexible, rubber-like parts directly from a digital file for the first time.

Processes include:

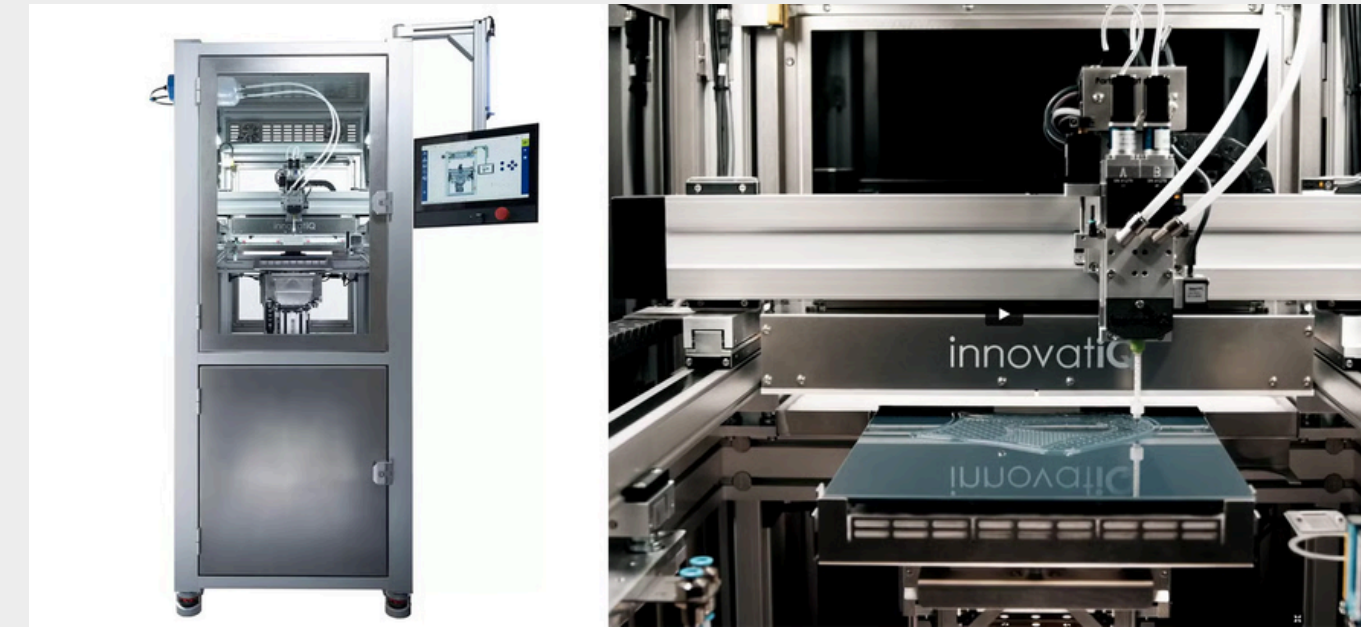
- Liquid Deposition Modeling (LDM)
- Drop-on-Demand (DoD) Printing

LIQUID DEPOSITON MODELING

This method, also known as Liquid Additive Manufacturing (LAM), functions by extruding a thick, paste-like silicone compound. The material is pushed through a nozzle and deposited onto a build platform, tracing the shape of each layer. The process is repeated, using a bottom up approach to build the part.

To solidify the material, heat is applied. Many systems use an integrated heating element or a halogen lamp. This heat triggers a chemical reaction within the silicone, causing it to cross-link and change from a paste into a solid, elastic rubber.

A leading example of this technology is the InnovatiQ LiQ 320 printer. This system uses a specialized Dow Chemical silicone formulation and can produce finished, functional parts. An optional coloring system allows for parts to be made in different colors during the printing process itself. This approach is well-suited for creating medical device parts with personalization.



LiQ 320 3D printer from German company InnovatiQ (Source: [InnovatiQ](#))

DOD PRINTING

Drop-on-Demand printing operates with high precision, similar to an inkjet printer but using silicone. The printhead releases individual, microscopic droplets of a liquid silicone mixture onto the build area. Each droplet is precisely placed to form a thin layer of the part.

After a layer is completed, a UV light passes over it. This light activates a special agent mixed into the silicone, starting the curing process and hardening the material. The printer then deposits the next layer on top.

A representative machine using this method is the Lynxter S300X. It features two independent extruders: one for printing the silicone part material and a separate one for printing a dissolvable support material, which is needed for creating complex overhanging geometries. This technology can print with a range of medical and industrial-grade silicones. It is important to note that it discontinued its development in 2021.



The Lynxter S300X silicone 3D printer (Source: [Lynxter](#))

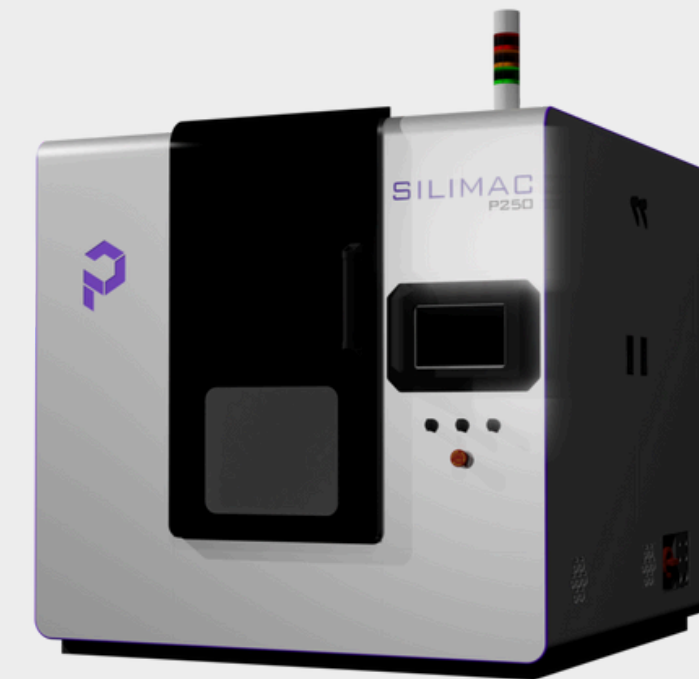
NEW PROCESSES

Beyond the established methods, new approaches are in development, aiming to solve remaining challenges. One interesting concept is gel-supported printing, shown by the new Rapid Liquid Print Levity system. This method proposes printing silicone directly inside a tank of a supportive gel. The gel holds the liquid silicone in place as it is deposited, eliminating the need for traditional support structures and potentially allowing for faster printing of large, flexible objects.

Approaches using highly specialized systems are emerging for fields like medicine. The Prayasta Silimac P250 is a printer designed specifically for manufacturing patient-specific soft-tissue implants and prostheses. It incorporates features like an in-built UV sterilization for print chamber and contamination-free support system, addressing the cleanliness requirements of implant manufacturing. It also has real-time curing using IR laser & heaters, increasing the time efficiency of the print.



The Rapid Liquid Print 's Levity (Source: [RLP](#))



Prayasta Silimac P250 (Source: [Prayasta](#))

CHALLENGES

Silicone is a unique elastomer that is not thermoplastic and resists UV curing. Unlike standard 3D printing materials, its low viscosity and inability to melt or harden via conventional means make it incompatible with common additive manufacturing processes:

- FDM Process: Relies on melting and extruding thermoplastics that solidify upon cooling, a mechanism that doesn't work for silicone, which doesn't melt in the same way.
- SLA Process: Requires UV-curable resins that harden quickly under light. Pure silicone cannot be cured this way without chemical modification, which often compromises its core properties like thermal stability and chemical resistance.

Even with modified chemical compositions, silicone's low viscosity and flexible nature present significant manufacturing hurdles:

- Dimensional Accuracy: Silicone tends to flow or spread unpredictably before curing, making it difficult to maintain clean layer definition and consistent geometry.
- Surface Finish: Achieving smooth, predictable surfaces is challenging compared to rigid materials.
- Support & Deformation: Printed parts are prone to deformation during both printing and post-processing, requiring careful support strategies and controlled curing environments.

SUPPLIERS

- Stratasys
- Formlabs
- Carbon
- B9 Creations
- Spectroplast
- Elkem Silicones
- Desktop Health
- Prayasta

**THANK
YOU**

Resources

- https://www.elkem.com/markets/advanced-manufacturing-industrial/molding-printing/3d-printing-additive-manufacturing/?st_source=ai_mode#:~:text=Elkem%20Silicones%20has%20developed%20a,Printing%20enhancing%20your%20competitive%20edge.
- https://lynxter.com/en/learn/blog/silicone-3d-printing-how-and-why-print-silicone?st_source=ai_mode
- <https://www.motiontech.ch/aboutus>
- https://solution.innovatiq.com/innovatiq/CustomUpload/3740357034003700356036903500326035603690364035203650337035603670337035203670/Brochure_LiQ320_english.pdf
- <https://www.innovatiq.com/en/drucker/sonstige>
- <https://www.rapidliquidprint.com/printer>
- <https://www.prayasta.com/ieam/>
- https://www.additive-x.com/blog/3d-silicone-printing-how-it-works-ideal-applications?srsId=AfmBOoplkaoJuyQowxCALDzys9x1i3l4QVctPVQd_vRuk5cx2zomUoP0
- <https://www.stratasys.com/en/resources/blog/silicone-3d-printing/#:~:text=Unlike%20many%20thermoplastics%20or%20rubbers,often%20compromises%20its%20core%20properties.>
- <https://investors.stratasys.com/news-events/press-releases/detail/948/stratasys-and-shin-etsu-launch-p3-silicone-25a-for>